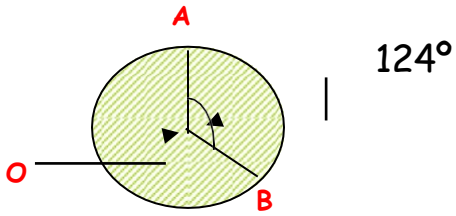
Mathematics Revision Test 36

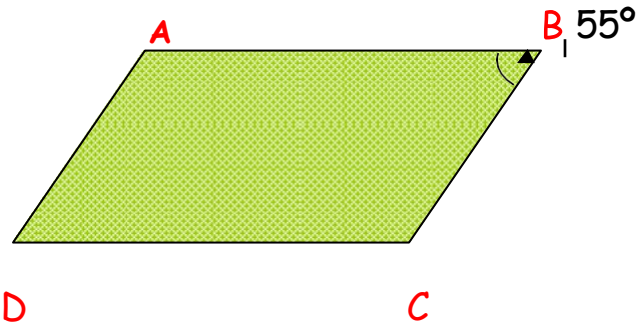
1. 0.84 =_____ thousandths

2. Write in words the number 1,000,000:_____

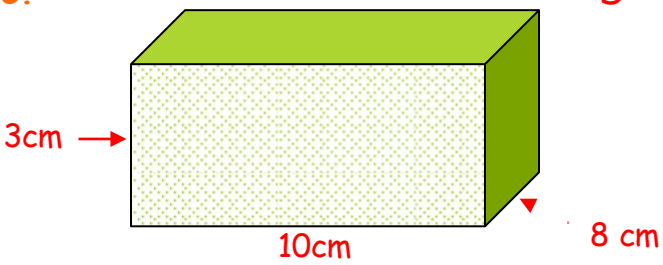
3. Find the **reflex** angle AOB:_____



4. ABCD is a parallelogram and angle ABC is 55°
What is the size of:
Angle ADC=_____
Angle DAB=_____
Angle BCD=_____



5.



a) How many cm cubes are needed to fill the box?

_____

b) If the box were **4 cm** high, find its volume in cm^3

_____

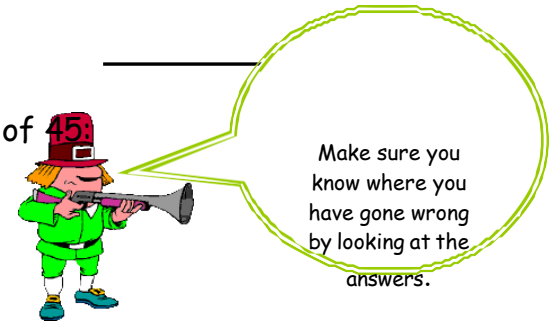
6. $\frac{4}{5}$ =_____ %

7. Write in figures half a million:_____

8. Find the PRODUCT of 5 and 16:_____

9. Find the difference between of $\frac{1}{5}$ 25 and $\frac{1}{3}$ of 45:

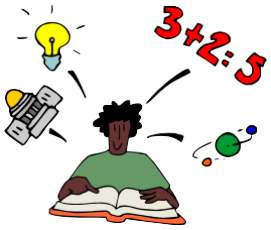
_____



10. 2.650 litres =_____ ml

How did you do? 100%? WOW!

NOT QUITE: Make sure you understand where you went wrong.



Calculation Ladders

You must be so fast by now!

These should take you just a few seconds

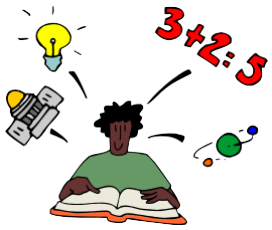
Ladder 18

1.	£18
2.	60% of this
3.	add on 20p
4.	X 9
5.	find $66\frac{2}{3}$ %
6.	10% of this
7.	find 50%
8.	10% of this
9.	add 67p
10.	Find 45%
	Your answer is

Ladder 19

1.	160
2.	find 20%
3.	50% of this
4.	$\sqrt{n}$
5.	75% of this
6.	n^2
7.	50% of this
8.	Divide by 3
9.	X 10
10.	Find 30%
	Your answer is

Are you STILL enjoying these?



Calculation Ladders

Now two more

These should take you even fewer seconds

Ladder 20

1.	£60
2.	60% of this
3.	plus £26
4.	find 50%
5.	add on £9
6.	Find 10%
7.	n^2
8.	add £4
9.	find 90%
10.	divide by 6
	Your answer is

Ladder 21

1.	£20
2.	find 20%
3.	add on 17
4.	$33\frac{1}{3}$ % of this
5.	double this
6.	multiply by 4
7.	10% of this
8.	40p extra
9.	X 8
10.	20% of this
	Your answer is

Are you STILL enjoying these?

Fraction ⟶ Decimal ⟶ Percentage

Quick check to make sure of these:

Vulgar Fraction (Lowest Terms)	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{3}{4}$	$\frac{1}{5}$	$\frac{2}{5}$	$\frac{3}{5}$	$\frac{4}{5}$	$\frac{1}{10}$	$\frac{3}{10}$	$\frac{7}{10}$	$\frac{9}{10}$	$\frac{1}{20}$	$\frac{1}{100}$
Decimal Fraction	0.5	0.25	0.75	0.2	0.4	0.6	0.8	0.1	0.3	0.7	0.9	0.05	0.01
Percentage	50%	25%	75%	20%	40%	60%	80%	10%	30%	70%	90%	5%	1%

I'm buzzing with excitement because I know these off by heart!

Addition and subtraction of fractions

Handy hints

You cannot add together fractions unless they have the same ***denominator*** (bottom number).

When your fractions have the same denominator, you simply add them together, or subtract.

Example:

To add $\frac{1}{3}$ to $\frac{1}{9}$ you must give them the same denominator.

$\frac{1}{3}$ is the same as $\frac{3}{9}$ So: $\frac{3}{9} + \frac{1}{9} = \frac{4}{9}$

Same denominators

The same rule applies to subtracting fractions:

To subtract $\frac{3}{1}$ from $\frac{2}{3}$ you must give them the same denominator.

$\frac{2}{3}$ is the same as $\frac{8}{1}$ So: $\frac{8}{1} - \frac{3}{12} = \frac{5}{12}$

Same denominators

Now complete the attached sheet, adding and subtracting fractions.

Do each sum like the examples
and you will get them all right!

These are really easy and good fun, so enjoy yourself!

Addition and subtraction of fractions

Leave your answers in lowest terms

$\frac{2}{12} + \frac{1}{4} =$ $\frac{2}{3} + \frac{1}{4} =$ $\frac{3}{5} + \frac{1}{4} =$ $\frac{2}{9} + \frac{1}{6} =$

$\frac{1}{3} + \frac{2}{12} =$ $\frac{1}{3} + \frac{2}{4} =$ $\frac{1}{5} + \frac{3}{20} =$ $\frac{4}{9} + \frac{1}{3} =$

$\frac{2}{3} + \frac{2}{12} =$ $\frac{1}{4} + \frac{1}{3} =$ $\frac{2}{3} + \frac{1}{9} =$ $\frac{1}{18} + \frac{5}{6} =$

$\frac{8}{12} + \frac{1}{4} =$ $\frac{1}{3} + \frac{1}{9} =$ $\frac{1}{6} + \frac{1}{8} =$ $\frac{1}{7} + \frac{1}{2} =$

$\frac{1}{3} + \frac{5}{12} =$ $\frac{2}{9} + \frac{2}{3} =$ $\frac{3}{8} + \frac{1}{3} =$ $\frac{1}{4} + \frac{3}{8} =$

$\frac{1}{6} + \frac{7}{12} =$ $\frac{1}{3} + \frac{7}{12} =$ $\frac{1}{4} + \frac{5}{8} =$ $\frac{2}{3} + \frac{1}{5} =$

$\frac{5}{6} + \frac{1}{12} =$ $\frac{1}{2} + \frac{1}{5} =$ $\frac{1}{3} + \frac{1}{5} =$ $\frac{1}{14} + \frac{3}{7} =$

$\frac{2}{3} - \frac{2}{12} =$ $\frac{1}{3} - \frac{1}{4} =$ $\frac{2}{3} - \frac{1}{9} =$ $\frac{5}{6} - \frac{1}{18} =$

$\frac{8}{12} - \frac{1}{4} =$ $\frac{1}{3} - \frac{1}{9} =$ $\frac{1}{6} - \frac{1}{8} =$ $\frac{1}{2} - \frac{1}{7} =$

$\frac{5}{12} - \frac{1}{3} =$ $\frac{2}{3} - \frac{2}{9} =$ $\frac{17}{24} - \frac{1}{3} =$ $\frac{3}{8} - \frac{1}{4} =$

$\frac{7}{12} - \frac{1}{6} =$ $\frac{2}{2} - \frac{6}{6} =$ $\frac{7}{12} - \frac{1}{3} =$ $\frac{10}{10} - \frac{4}{4} =$

$\frac{5}{6} - \frac{1}{12} =$ $\frac{1}{2} - \frac{1}{5} =$

$\frac{2}{3} - \frac{3}{12} =$ $\frac{1}{2} - \frac{1}{10} =$

$\frac{1}{1} - \frac{2}{2} =$ $\frac{3}{3} - \frac{1}{1} =$

$\frac{5}{8}$ $\frac{1}{4}$

$\frac{1}{3}$ $\frac{1}{5}$

$\frac{2}{5}$ $\frac{3}{10}$

$\frac{7}{10}$ $\frac{1}{5}$

$\frac{2}{3}$ $\frac{2}{5}$ = - = -

=

$\frac{3}{7}$ $\frac{1}{14}$ = - = -

= = -

$\frac{5}{8}$ $\frac{1}{16}$ = - = -

=

$\frac{1}{4}$ $\frac{3}{16}$ = -

=

Have you got them all correct?

11 PLUS PREPARATION

MATHS PAPER

Name:	
Date of Birth:	
Today's Date:	

READ THESE INSTRUCTIONS CAREFULLY BEFORE STARTING.

- You have 35 minutes to complete the paper.
- Working out can be done on the paper or on spare paper but ensure only your answer is written clearly on the answer line.
- Each question is worth one mark. If you can't do a question, then move on.
- If you finish early then check your answers.
- Once the test has started you may not ask for help.
- No calculators should be used.

1

1. **Calculate:** 4/7 of 63 _____

---

2. **Work out the value of:**

   12 – (8-3) x 2 _____

---

3. **Calculate:** 4000 – 747 _____

---

4. **Calculate:** 462 + 369 _____

---

5. What is the difference between the largest and the smallest of the following numbers? _____

   A. 0.85 **B.** 0.49 **C.** 0.9 **D.** 0.79 **E.** 0.76

---

6. Which of these solid shapes has exactly four faces? _____

A Cube

B Cuboid

C Cylinder

D Tetrahedron

E Triangular Prism

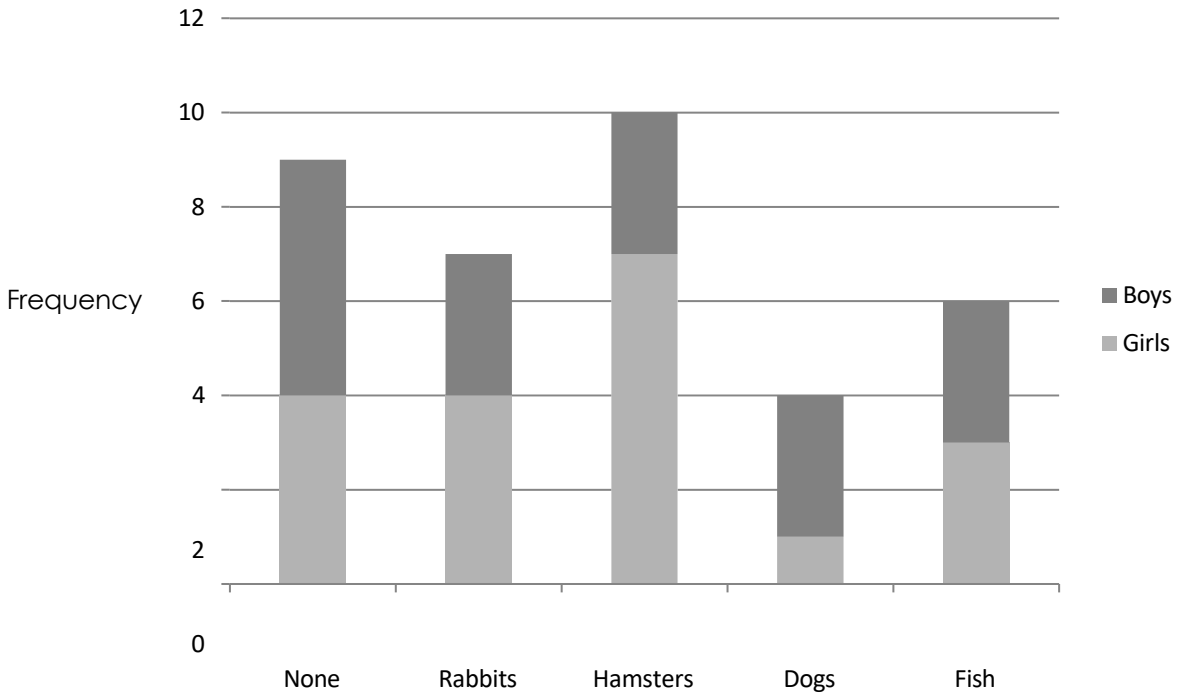
---

7. Helen bought 7 melons and was given 87p change from £12.00. How much did each melon cost? _____

8. A litre bottle of fizzy drink is completely poured into 8 identical glasses. If each glass contains the same amount of drink how many millilitres of fizzy drink will be in each glass? _____

---

9. The bar chart gives information about the pets owned by the children that live in one road.



a/ How many boys do not have a pet? _____

b/ What percentage of the dog owners are girls? _____

c/ If the number of children living in the street is 28, how many own at least one pet? _____

---

10. Peter gets £5 a week pocket money. His friends John, Phillip and George get £10.50, £12.00 and £8.30 each. What is the average (mean) amount of pocket money received by the four boys? _____

---

11. Four hot dogs cost £10. How much will seven hot dogs cost? _____

2

12. A pub hires out glasses for a deposit (d) plus a small charge for each glass. The total cost (C) can be worked out using this formula:

C = 0.1n + d (n is the number of glasses)

If fewer than 50 glasses are hired then d = £4

If 50 or more glasses are hired then d = £8

How much will Jamie pay to hire 75 glasses? _____

---

13. Peter was born 23rd August 1995. How old was he in **complete** years and months on 25th December 2010? _____

---

14. Which row shows a multiple of 4, a prime number and a square number in that order? _____

A/	**20**	**9**	**16**
B/	**55**	**23**	**24**
C/	**24**	**25**	**9**
D/	**28**	**19**	**64**
E/	**100**	**29**	**50**

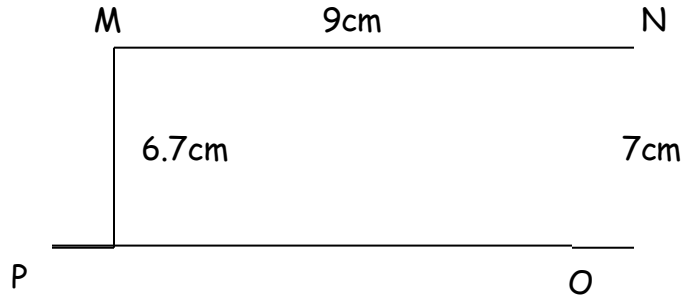
15. 14 winning tickets shared a lottery prize of £10.5 million (£10,500,000). How much prize money will each owner of a winning ticket receive? _____

---

16. In a test the pass level is 80%. Is a score of 19 correct answers out of 25 questions a pass or a fail? _____

---

17. What is the area of the parallelogram MNOP? _____



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18. Jane draws a plan of her house using a scale of 3cm to 8m. On the plan the front of the house is 6cm long. What is the real length of the front of the house? _____

---

19. The train was due to arrive at St. Pancras station at 9.25am but was delayed by 37 minutes. How many minutes did the passenger have left to walk from the train to their 10.30 am meeting? _____

---

20. What is the perimeter of this shape? _____

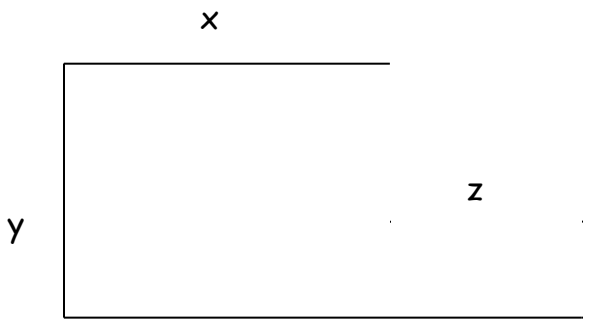
A/ 2x + y + z

B/ x + y + z

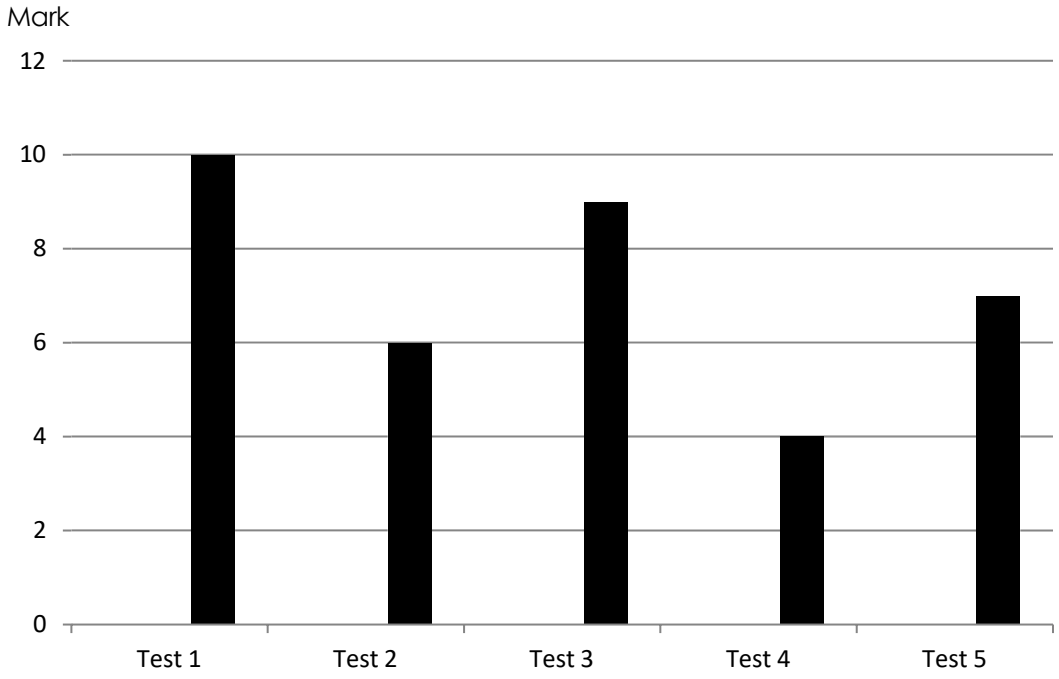
C/ x + 2 (y+z)

D/ 2 (y+z)

E/ 2 (x + y + z)

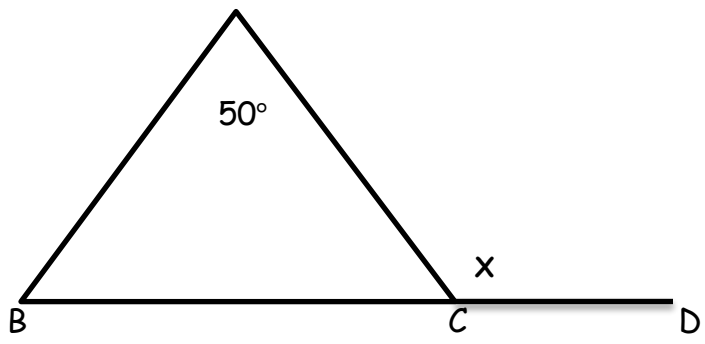


21. The graph shows Tom's 5 English test results. What is Tom's average (mean) score for English? _____



22. The triangle in the picture is isosceles; sides AB and AC are equal.

How large is the angle at x? _____



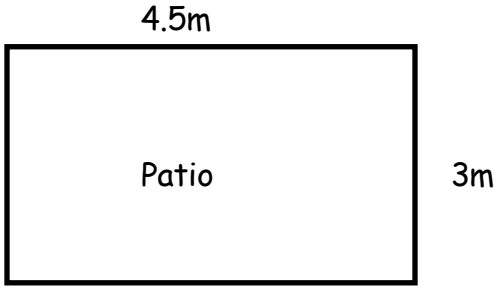
5

23. An isosceles triangle has one angle of 30°

What do each of the other angles measure? _____

---

24. Janet decides to lay Old York stone on her patio. Each slab of stone measures 50cm x 50cm.



The stone slabs are sold in lots of 12.

A/ What is the area of the patio in m^2? _____

B/ How many stone slabs are needed to cover the patio exactly? _____

C/ How many full lots of stone slabs should Janet buy? _____

---

25. The formula for converting temperatures in degrees Fahrenheit to degrees Centigrade is:

$$C = 5 \times (F-32) \div 9$$

What is the value of C when F is:

a/ 40 _____

b/ 8 _____

and write your answer to two decimal places.

END OF TEST

If you have time

- **Go back and complete any questions you missed out.**
- **Check your answers carefully.**

ANSWERS - COMPLETE MATHS – PART 38

Maths

Revision Sheet

1. 840

1. one million
2. 236°
3. 55° 125° 125°
4. a. 240 b. 320cm^3
5. 80%
6. 500,000
7. 80
8. 10
9. 2650ml

Calculation Ladders

Sheet 1

Ladder 18 = 45p

Ladder 19 = 4.5

Sheet 2

Ladder 20 = £3

Ladder 21 = £9.60

Fraction addition and subtraction

(Answers from left to right, top to bottom)

1 $\frac{5}{12}$

2 $\frac{11}{12}$

3 $\frac{17}{20}$

4 $\frac{7}{18}$

5. $\frac{1}{2}$

6. $\frac{5}{6}$

7 $\frac{7}{20}$

8. $\frac{7}{9}$

9. $\frac{5}{6}$

10. $\frac{7}{1}$

11. $\frac{7}{9}$

12. $\frac{8}{9}$

13. $\frac{11}{1}$

14. $\frac{4}{9}$

15. $\frac{7}{2}$

16. $\frac{9}{1}$

17. $\frac{3}{4}$

18. $\frac{8}{9}$

19. $\frac{17}{2}$

20. $\frac{5}{8}$

21. $\frac{3}{4}$

22. $\frac{11}{1}$

23. $\frac{7}{8}$

ANSWERS - COMPLETE MATHS – PART 38

24. $\frac{13}{1}$
25. $\frac{11}{1}$
26. $\frac{7}{1}$
27. $\frac{8}{1}$
28. $\frac{1}{2}$
29. $\frac{1}{2}$
30. $\frac{1}{1}$
31. $\frac{5}{9}$
32. $\frac{7}{9}$
33. $\frac{5}{1}$
34. $\frac{2}{9}$
35. $\frac{1}{2}$
36. $\frac{5}{1}$
37. $\frac{1}{1}$
38. $\frac{4}{9}$
39. $\frac{3}{8}$
40. $\frac{1}{8}$
41. $\frac{5}{1}$
42. $\frac{1}{4}$
43. $\frac{3}{8}$
44. $\frac{4}{1}$
45. $\frac{3}{4}$
46. $\frac{3}{1}$
47. $\frac{2}{1}$
48. $\frac{5}{1}$
49. $\frac{5}{1}$
50. $\frac{2}{5}$
51. $\frac{1}{1}$
52. $\frac{9}{1}$
53. $\frac{1}{6}$
54. $\frac{1}{2}$
55. $\frac{1}{2}$
56. $\frac{1}{1}$

Maths Type 2 Paper 2

<u>Page 1</u>

1. 36
2. 2
3. 3253
4. 831

ANSWERS - COMPLETE MATHS – PART 38

5. 0.41
6. D
7. £1.59

Page 2

8. 125ml
9. a. 5
   b. 25%
   c. 19
10. £8.95
11. £17.50

Page 3

12. £15.50
13. 15 years 4 months
14. D
15. £750,000
16. Fail

Page 4

17. 60.3cm ^{2}
18. 16m
19. 28 min
20. E

Page 5

21. 7.2
22. 115°

Page 6

23. 75°
24. a. 13.5m ^{2}
    b. 54
    c. 5
25. a. 4.44
    b. -13.33